Implementation and Validation of 1D Return Mapping Algorithms for Plasticity and Viscoplasticity

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Abstract

Constitutive material models are the cornerstone of finite element analysis in solid mechanics. Material behavior defines how displacements, velocities and accelerations are calculated at the nodes of each elements. For viscoplastic behavior, mathematical definitions ties the evolution of the viscoplastic strain to the evolution of stresses and strain over time. Thus for each increment of strain and time, algorithm must find the equilibrium at wich Karush-Kuhn-Tucker (KKT) conditions are met. However, since viscoplastic material exhibit stress dissipation, these conditions needs to be relaxed to accomodate the accumulation of viscoplastic strain. In this mock paper, a return mapping algorithm for the Perzyna viscoplastic model will be implemented to solve for the equilibirum of the KKT conditions considering the overstress concepts. The basic definitions of return mapping for regular plasticity will be showed and will serve as a basis for further implementation of the Perzyna model to show the impact of strain rates and material properties on constitutive behavior.

Keywords: Return mapping, Plasticity, Viscoplasticity, Perzyna model, Backward Euler

1. Introduction

Consitutive material models are required to model the material response to loads in finite element analysis. For solid mechancis, variable of interest are the displacement, veolicity and acceleration of each and every node to compute variable of interest like stresses and strains. For linear behavior, where the stress response follows a linear relationship with strains (by definition displacements) this problem is rather easy to solve but even then, advanced algorithm are used to determine the displacements, stresses and strains efficiently. The same goes for other material behavior like plasticity and viscoplasticity.

Return mapping is a well known algorithm and have been used for well over forty years. Early work by [**?**] on radial return have pave the way to the work of [**?**]. Since then, great textbook have also covered the topic and applied it to several material models (consitutive model)[**?**] [**?**].

The goal of this mock paper is to condense gathered knowledge about computational plasticity, especially about return mapping applied to 1D viscoplastic constitutive Perzyna model. First, a survey of the theoretical background of return mapping will be presented. Then, The implementation of the algorithms will be shown and finally key results and observation to common material will be studied.

2. Theoritical background

The governing equations in solid mechanics are defined by:

$$\frac{\partial \sigma_{ij}}{x_j} + \rho b_i = \rho \ddot{u} \tag{1}$$

$$\sigma_{ij} = C_{ijkl}\varepsilon_{ij} \tag{2}$$

$$\varepsilon_{ij} = \frac{1}{2}\left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i}\right) \tag{3}$$

known as **??**: the equilibrium equation , **??**: the constitutive law and **??**: the strain-displacement relation.

In a purely linear, 1D isotropic case (2) can be simply seen as the product of the Young's modulus (E) and the strain applied to a rod (ε). In multidimensional nonlinear computational mechanics, this can also be seen as the *trial Stress* or *elastic predictor*. As introduced in the last sentece, return mapping starts with an initial guess to see if the new increment of strain generate enough stress so that plasticity or viscoplasticity is indused in the material. To consider that inelastic stress is induced, the *elastic predictor* must give a value that is beyond elastic stress, thus a value larger than σ_y. The stress from a loading is given by a yield function (e.g. Von Mises), and the concept of if the stress state is elastic or inelastic can be verified with a simple concept of yield criterion (f). Then, three conditions has to be met at all times, the Karush-Kuhn-Tucker (KKT) conditions:

$$\gamma \geq 0 \tag{4}$$

$$f(\sigma) \leq 0 \tag{5}$$

$$\gamma f(\sigma) = 0 \tag{6}$$

Here, the symbole γ is called the plastic multiplier and is part of the return mapping algorithm. Essentially, the plastic multiplier scale the equivalent inelastic strain in the current increment. And simply, equations **??**, **??**, **??** constrain the problem

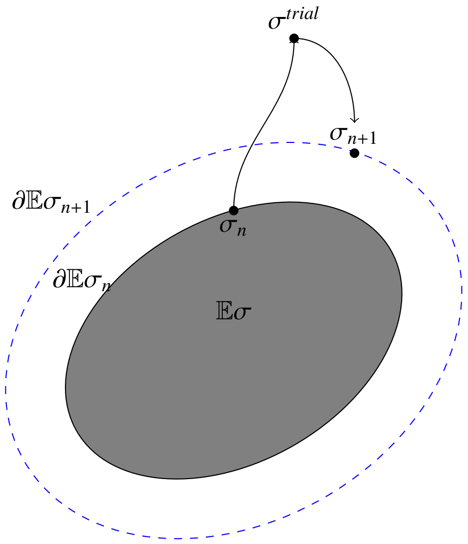

Figure 1: Return mapping in 1D with strain hardening

so that for any stress state either $f(\sigma) < 0$ meaning that the stress state is purely elastic or either $\gamma > 0$ meining inelastic strain is increasing. One other important aspect is that $f(\sigma)$ must at most be equal to zero. To better visualize, see this figure:

As seen in **??**, an inital stress state at step (n) is defined by a domain of admissible stress $(\mathbb{E}\sigma)$ and a boundary to this domain $(\partial\mathbb{E}\sigma_n)$ **** AJOUTER VECTEUR DE LA SURFACE À SIGMAN+1 *****

2.1. Rate-Independent Plasticity

- Yield function: $f(\sigma) = |\sigma| - \sigma_y$
- Flow rule with normality
- Kuhn-Tucker conditions
- Key equation: $\Delta\lambda = f^{trial}/E$

2.2. Viscoplasticity

In some cases and conditions, metals exhibit viscoplastic behavior. Viscoplasticity can be defined by the dependence of meterial state evolution to the strain rate.

- Perzyna mode: $\dot{\varepsilon}^{vp} = \frac{1}{n}\left\langle \frac{f}{\sigma_y} \right\rangle^n sign(\sigma)$
- Physical meaning of η and n
- Limiting case $n \to 0$ (recovers rate indenpendence)

3. Implementation

3.1. Algorithm Structure

- Elastic predictor
- Yield check
- Plastic corrector
- Figure
  - Simple flowchart to illustrate the process of return mapping

3.2. Code Organization

Algorithm 1: Max finds the maximum number

Input: Current step material data $(\sigma_n, \varepsilon_n, \varepsilon_n^p)$
Output: Updated material data $(\sigma_{n+1}, \varepsilon_{n+1}, \varepsilon_{n+1}^p)$
$max \leftarrow a_1$
for $i \leftarrow 2$ ***to*** n **do**
  if $a_i > max$ **then**
    $max \leftarrow a_i$
  end
end
return max

Appendix A. Example Appendix Section

Appendix text.

References

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1	2	3
4	5	6
7	8	9

Table A.1: Table Caption